User

Command interface
Parse commands and read/write files

publish, install, show

Package registry

Publish / install / show
Sign publications and resolve versions

Validation
Check ownership, signatures, and history
Reject forks and rollback; fetch missing history

put, get, ping

store and retrieve

validate registry updates

owner public key

upload and download
follow history
resolve and create

Ownership verifier
DNS TXT or fake DNS

Package data

Latest pointer
Name-derived key; signed mutable value

record hash

Version records
Signed records linked to previous versions

previous record hash

blob hash

Package blobs
Immutable bytes addressed by content hash

Kademlia DHT

Join / lookup / store / replicate
Find closest peers and distribute values

select and learn peers

read and write

Routing table
Contacts in k-buckets; XOR distance

Local storage
Content-addressed values and validated latest pointers

requests and results

RPC layer
JSON requests and replies
Request IDs, response channels, timeouts

Network transport

SimulatedNode
In-memory message channels

UDPNode
UDP datagrams

same process

processes or containers

Other nodes
Same registry, DHT, and storage